AYRA AGRAWAL

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Education

Rutgers University – New Brunswick Expected May 2029
B.S. Computer Science | B.A. Psychology | Minor: Data Science | GPA: 3.6/4.0 New Brunswick, NJ
Relevant Coursework: Data Structures, Computer Architecture, Discrete Structures I, Linear Algebra (in progress)

Projects

TraceLab – Distributed Tracing & Observability Backend | *GitHub*
Python, FastAPI, PostgreSQL, Redis, Docker, AWS EC2, Prometheus, Locust

- Designed an asynchronous ingestion path: the FastAPI endpoint validates each trace, enqueues it to Redis, and returns 202, while an independent worker persists traces to PostgreSQL and drops duplicates via unique constraints.
- Modeled parent-child span trees rejecting missing or cross-trace parents; added paged queries and Redis-cached lookups; deployed 4 containers to AWS EC2 with Prometheus metrics and 11 Pytest tests in CI.
- Load-tested enqueue locally (Locust, 100 users): **484 req/s avg, 96 ms p95, 0% HTTP failures**; DB writes excluded.

Recall – Full-Stack AI Research Workspace | *GitHub* | *Demo (controlled, recorded run)*
Python, FastAPI, PostgreSQL/pgvector, Sentence Transformers, Ollama, Next.js, TypeScript

- Built an end-to-end research workflow: PDFs are parsed with PyMuPDF into overlapping chunks, embedded (384-dim) into pgvector, and answered with page-level citations that open the source page in a Next.js PDF viewer.
- Designed hybrid retrieval that merges pgvector and query-expanded BM25 candidates and reranks them with a cross-encoder; wrote evaluation scripts scoring page hits, recall@k, and abstention against a vector-only baseline.
- Streamed local Ollama LLM answers through FastAPI; 39 Pytest tests cover upload, query, streaming, and persistence.

SpotOn – Computer Vision Fitness Analyzer | HackRU Fall 2026, 3-person team | *GitHub* | *Demo*
TypeScript, Next.js, React, MediaPipe, Gemini API, Zod, Vitest

- Built the Gemini analysis route and results UI: Zod request validation, JSON-schema-constrained output grounded in measured rep metrics, a concurrency cap returning HTTP 429, and Vitest tests.
- Helped refine the team's MediaPipe pipeline: calibration, completed-rep counting, replay, and set comparison.

Experience

Undergraduate Research Assistant & IT Team Member May 2026 – Present
Cognition and Learning Center (CALC), Rutgers Center for Cognitive Science New Brunswick, NJ

- Engineered R data pipelines that parse **227+ behavioral coding files**, extract measurement periods and frame onsets, and compute looking durations into structured, analysis-ready datasets for an infant cognition study.
- Built automated validation into the pipeline that flags inconsistent coding records before statistical analysis.
- Developing EyeSpy (ongoing), a PsychoPy number-word experiment; configure Tobii Pro eye-trackers with DeTox.

Backend Developer Sep 2026 – Present
Hack4Impact – Rutgers Chapter New Brunswick, NJ

- Own authentication for Love Blue Inc.'s volunteer-management platform, a Supabase app the team is building to connect volunteers, local chapters, cleanup events, and environmental-impact tracking.
- Implementing Supabase Auth sign-in providers and automatic profile creation on signup, the identity layer the team's planned role-aware access for volunteers, chapter leaders, and national admins will rely on.

Frontend Software Engineering Fellow | Rutgers Blueprint Oct 2026 – Present

- Completed HTML/CSS and JavaScript/DOM in an 8-week fellowship; TypeScript, React, and a team project are next.

Backend Developer Feb 2026 – May 2026
Rutgers Mobile App Development (RUMAD) Incubator New Brunswick, NJ

- Wrote RUThrift's Express/TypeScript auth API: session-checked email OTP verification and Supabase profile upserts.
- Chose email OTP over SMTP link verification to balance onboarding speed and security, serving a 4-person team's React Native/Expo marketplace app; presented at the RUMAD showcase (50+ attendees).

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C, SQL, R, HTML/CSS
Frameworks & Databases: FastAPI, Express, React, Next.js, React Native, PostgreSQL, pgvector, Redis, Supabase
Infrastructure & Testing: Docker, AWS EC2, GitHub Actions, Prometheus, Git, Pytest, Vitest, Locust
AI & Research Tools: Sentence Transformers, Ollama, MediaPipe, Gemini API, PsychoPy, Tobii Pro, DeTox, Datavyu
Certifications & Awards: Google Cybersecurity Certificate; Duke of Edinburgh's International Award